MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC, 5-1/2 Inch (Kocher)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers curved and straight hemostatic forceps of the Kocher design.

1.2 Classification. Forceps shall be of the following types as indicated below (see 6.2).

Type

1. Straight
2. Curved

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

24439 - FORCEPS, HEMOSTATIC, KOCHER

(Copies of specifications and standards, and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

F 921 - Standard Definitions of Terms Relating to Hemostatic Forceps.

F 1026 - General Workmanship and Performance Measurements of Hemostatic Forceps.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The hemostatic forceps shall be in accordance with the applicable requirements specified herein. The terms used to describe the components and portions of the instruments described in this specification shall be in accordance with ASTM F921.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

3.2 Material. The forceps shall be fabricated from Class 4, martensitic stainless steel in accordance with ASTM F899.

3.3 Hardness. The forceps shall have a Rockwell hardness of C40 to C47 when tested as specified in 4.4.1. The requirements of GG-I-526 shall apply.

3.4 Style, design and dimensions. The forceps shall be of the Kocher design with either straight or curved jaws suitable for general surgical procedures. The distal tip of the jaw shall have a single longitudinally oriented tooth on one jaw and a mating groove on the other. The jaws shall be consistent in depth and in accordance with GG-I-526. The forceps shall mesh completely and properly during closure. The action of the forceps shall be easy, without drag or bending action. The forceps shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing #24439.

3.5 Locks. Shall be as specified in ASTM F1026.

3.6 Finish. Finish shall be entirely mirror, satin or matte. Should the instrument be of mirror finish, the locks and ratchets shall be polished smooth, but need not possess a mirror finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.7 Passivation. The forceps shall be properly passivated.

3.8 Jaw and ratchet operation. Shall be tested as specified in ASTM F1026. Forceps which do not perform as specified shall be rejected.

3.9 Identification marking. In addition to the requirements specified in ASTM F1026, forceps of foreign manufacture shall be marked "stainless."

3.10 Workmanship. The forceps shall be free from burrs, cracks, pits, sharp edges, scale, corrosion, or other defects which detract from its appearance or impair its serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion-resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document not met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one forceps.

TABLE I. Sampling criteria for inspection

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
Dimensions	3.4	---	S-2	2.5
Distortion test	ASTM F1026		S-2	1.0
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1.0
Jaw and ratchet operation test	3.8 & ASTM F1026		S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

4.3 Examination.

4.3.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Forceps not of the specified design for appropriate type, stock number and item identification.
102	Both jaws not serrated.
103	Jaw serrations not of uniform depth and not well defined.
104	Jaw serrations not chamfered throughout full length.
105	Ratchet teeth not well defined.
106	Forceps not free from cracks, pits, burs, sharp edges, scale, rough areas or corrosion.
107	Finish of forceps not uniform and not as specified.
108	Serrations of both jaws do not completely mesh in fully closed position.
109	Box lock performance not as specified in 3.5.
110	Jaws possess sharp tips.
111	Ratchet teeth do not completely engage in all positions.
112	Pin for box lock not flush with surface and not free from crevices.
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Forceps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each forceps shall be packaged in accordance with MIL-P-116, method IC-3.

5.1.1.4 Intermediate package. Twelve unit packages containing forceps shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of box shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for forceps shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages containing forceps of one type only shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

inserted into an optical comparator, shadowgraph, or other suitable

5.2.1 Level A. Twelve intermediate packages (144 units) containing forceps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (144 units) containing forceps of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Twelve intermediate packages (144 units) containing forceps of one type only shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Forceps packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Forceps packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The hemostatic forceps shall be used for general surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National Stock Number.

6.3 National Stock Number Coverage. This specification covers the following National Stock Numbers:

National Stock No.	Item Identification	Specification Destination
6515-01-138-5014	FORCEPS, HEMOSTATIC, Straight, Kocher, 5-1/2 Inch	Type 1
6515-01-138-5015	FORCEPS, HEMOSTATIC, Curved, Kocher, 5-1/2 Inch	Type 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

6.4 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistant than satin finish.

6.5 Subject term (key word) listing.

Forceps
Hemostatic forceps
Kocher forceps

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3870

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:17Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
Attn: DPSC-RSTS
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:17Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-42045	FORCEPS, HEMOSTATIC, 5-1/2 INCH (Kocher)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 20260-21T20:17Z
Check the source to verify that this is the current version before use.